

SYS 304

Trade Studies, Risk, and Decision Analysis

Advanced Trade Studies and Decision Analysis Methods

Quality Function Deployment

Quality Function Deployment

- ❑ Introduced by Yoji Akao, 1966.
 - Kobe Shipyard of Mitsubishi Heavy Industries, Ltd., Japan.
 - Used in manufacturing, heavy industry, and systems engineering.
 - Also applied to software engineering by IBM, HP, Texas Instruments, etc.
- ❑ **Supports information transformation.**
 - Customer requirements into product attributes.
- ❑ Mechanism can be extended and deployed successively across multiple levels of system decomposition and definition.

Quality Function Deployment

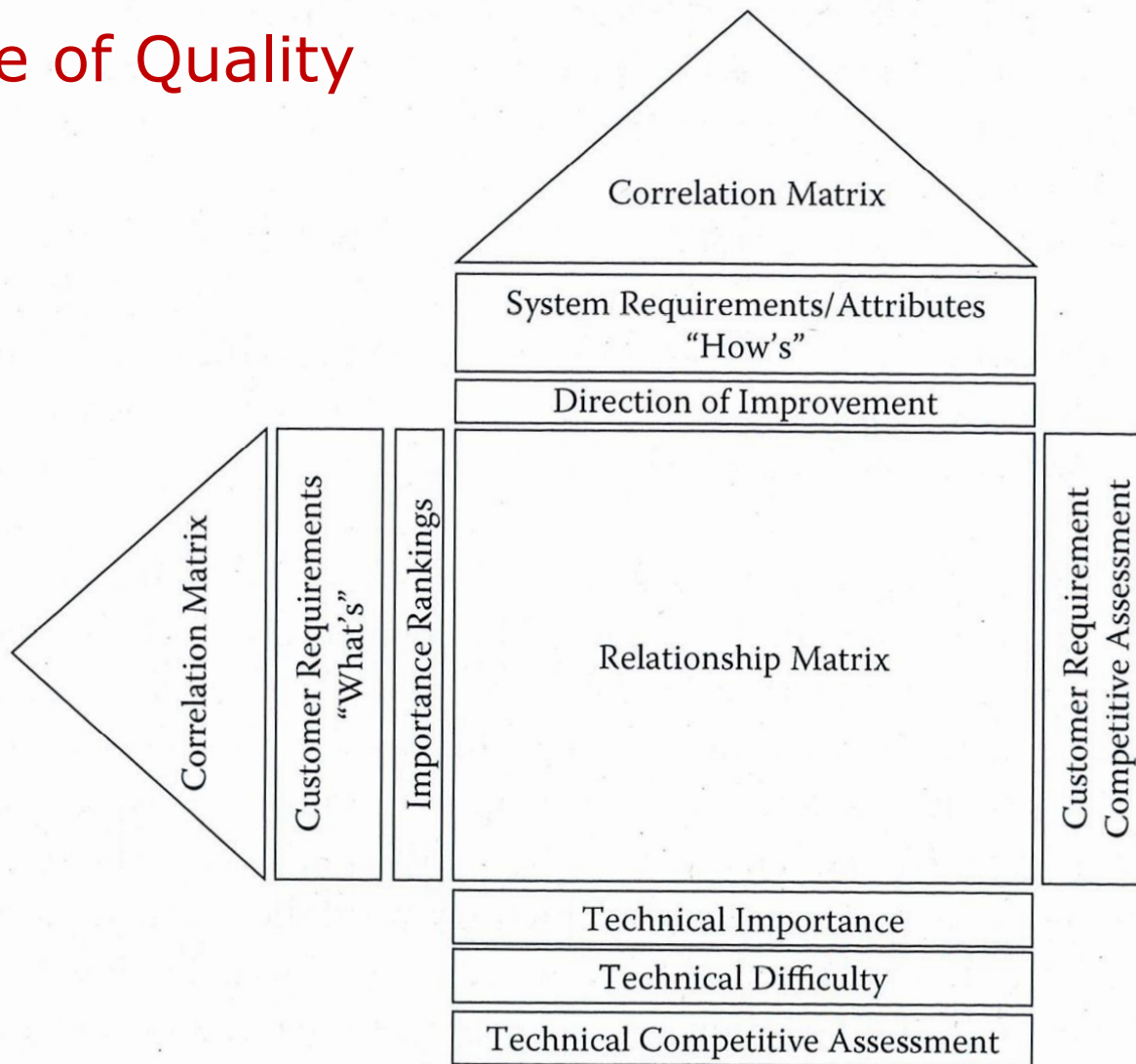
- ❑ Technique for discovering customer requirements and defining major quality assurance points to be used throughout the production phase.
 - Provides a structure for ensuring that customers' needs and desires are acknowledged and directly translated into technical requirements, from analysis through implementation to deployment.
- ❑ Constructs relationship matrices between customer needs, technical requirements, priorities, and (if needed) competitor assessment.

Quality Function Deployment

- ❑ Methodology for translating customer needs into specific system or product characteristics, and then for specifying the processes and tasks needed to produce that system or product.
 - Requires that the project team articulate the means by which the product or system designed will achieve customer requirements.
 - Process starts with market needs and then uses a planning matrix called the house of quality to translate the needs or requirements into the necessary technical requirements.

Graphical Representation

□ House of Quality

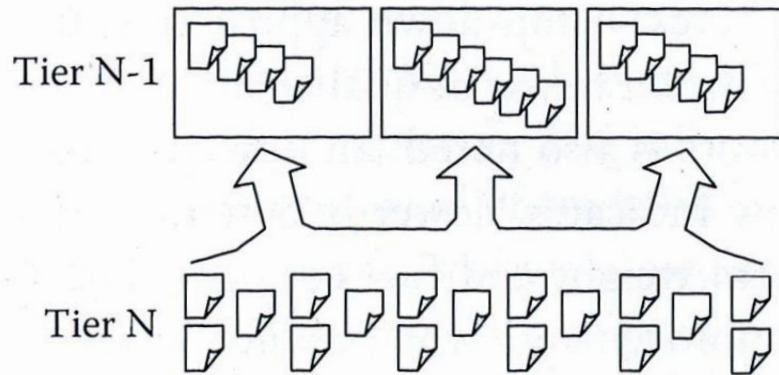


House of Quality

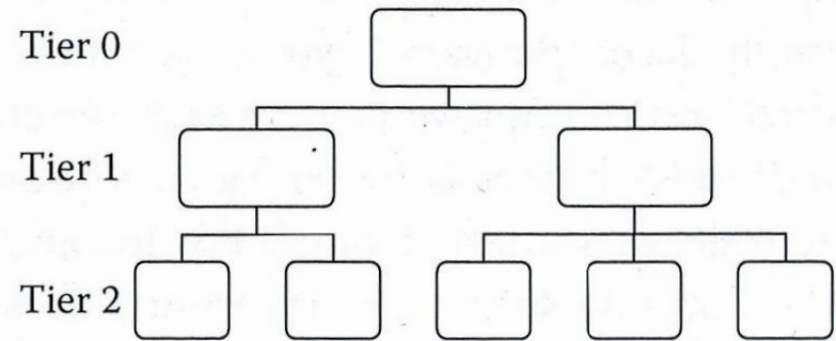
- ❑ Uses expert opinion to facilitate and document relevant information transformation.
 - Transformation of customer requirements (“What’s”) into product attributes (“How’s”).
- ❑ Requirements can be issued using a top-down or a bottom-up approach.
 - Tree diagram for top-down approach.
 - Affinity diagram for bottom-up approach.
 - Prioritization matrix to enable prioritization of requirements.

Supporting Tools for QFD

Affinity Diagram



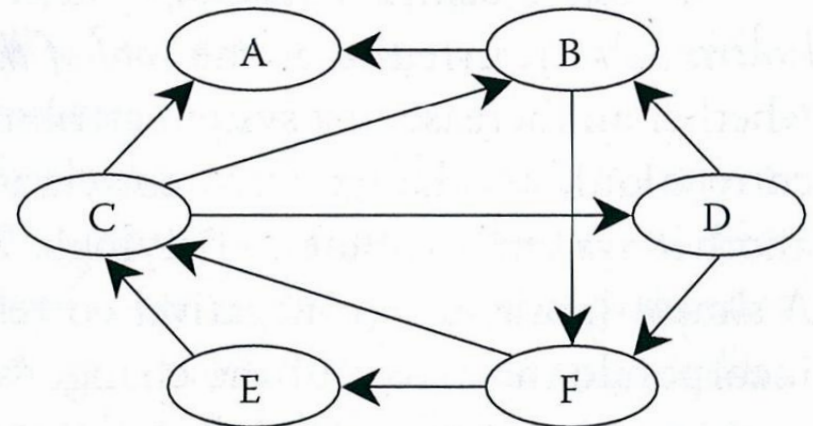
Tree Diagram



Prioritization Matrix

	Req. 1	Req. 2	Req. 3	Req. 4	Req. 5	...	Req. n	
Req. 1	1							$\rightarrow \Sigma_1 \rightarrow R_1 = \Sigma_1 / \Sigma_{TOTAL}$
Req. 2		1					1/3	$\rightarrow \Sigma_2 \rightarrow R_2$
Req. 3			1					
Req. 4				1				$\vdots$
Req. 5		3			1			
...								
Req. n							1	$\rightarrow \Sigma_n \rightarrow R_n$
								Σ_{TOTAL}

Interrelationship Diagram



House of Quality

❑ Customer Requirements: “What’s”

- Customer requirements competitive assessment.
- Tree diagram for top-down decomposition.
- Affinity diagram for bottom-up composition.

❑ Importance Ranking.

- Prioritization matrix to support prioritization of requirements.

❑ Correlation Matrix (Greenhouse).

- Measure correlation among customer requirements.
- Supports execution of trade-off studies in the design process.
- Can help in identifying conflicting requirements.
- Use of the greenhouse is optional.

House of Quality

❑ System Requirements: “How’s”

- Technical competitive assessment.
- Tree diagram for top-down decomposition.
- Affinity diagram for bottom-up composition.

❑ Technical Importance.

- Estimated for each system attribute as a weighted sum of all customer requirements relationship values, for that attribute, where the weights are the importance ratings for the customer requirements.

❑ Technical Difficulty.

- Design requirement assessed and documented for each system attribute with a discrete scale.
- Combination of technical importance and difficulty provides a measure of technical risk.

House of Quality

□ Direction of Improvement.

- An upward arrow indicates "higher is better"
- Downward arrow indicates "lower is better"
- Zero denoted a target value.

□ Correlation Matrix (Roof).

- Indicates whether an increase in a system attribute is accompanied by an increase (positive correlation), no change (zero correlation), or a decrease (negative correlation) on another system attribute.
 - {positive, zero, negative}
 - {strong positive, weak positive, zero, weak negative, strong negative}
- Generation supported by expert opinion or tools such as the interrelationship diagram.

House of Quality

Correlation Matrix.

	System Req. 1	System Req. 2	System Req. 3	System Req. 4	System Req. 5	⋮	System Req. n
System Req. 1		+	○	○	#	#	
System Req. 2	+		#	+	+	○	
System Req. 3	○	#		○	○	+	
System Req. 4	○	+	○		#	○	
System Req. 5	#	+	○	#		○	
⋮							
System Req. n	#	○	+	○	○		



	System Req. 1	System Req. 2	System Req. 3	System Req. 4	System Req. 5	⋮	System Req. n
System Req. 1		+	○	○	#	#	
System Req. 2	+		#	+	+	○	
System Req. 3	○	#		○	○	+	
System Req. 4	○	+	○		#	○	
System Req. 5	#	+	○	#		○	
⋮							
System Req. n	#	○	+	○	○		

Negative Correlation #

No/Weak Correlation ○

Positive Correlation +

House of Quality

❑ Relationship Matrix (Main Room).

- Captures the relational structure between the two sets of attributes.
- Importance of each system attribute relative to attaining the different customer requirements.
- More importance denotes a stronger relationship.

❑ Customer Requirements Competitive Assessment.

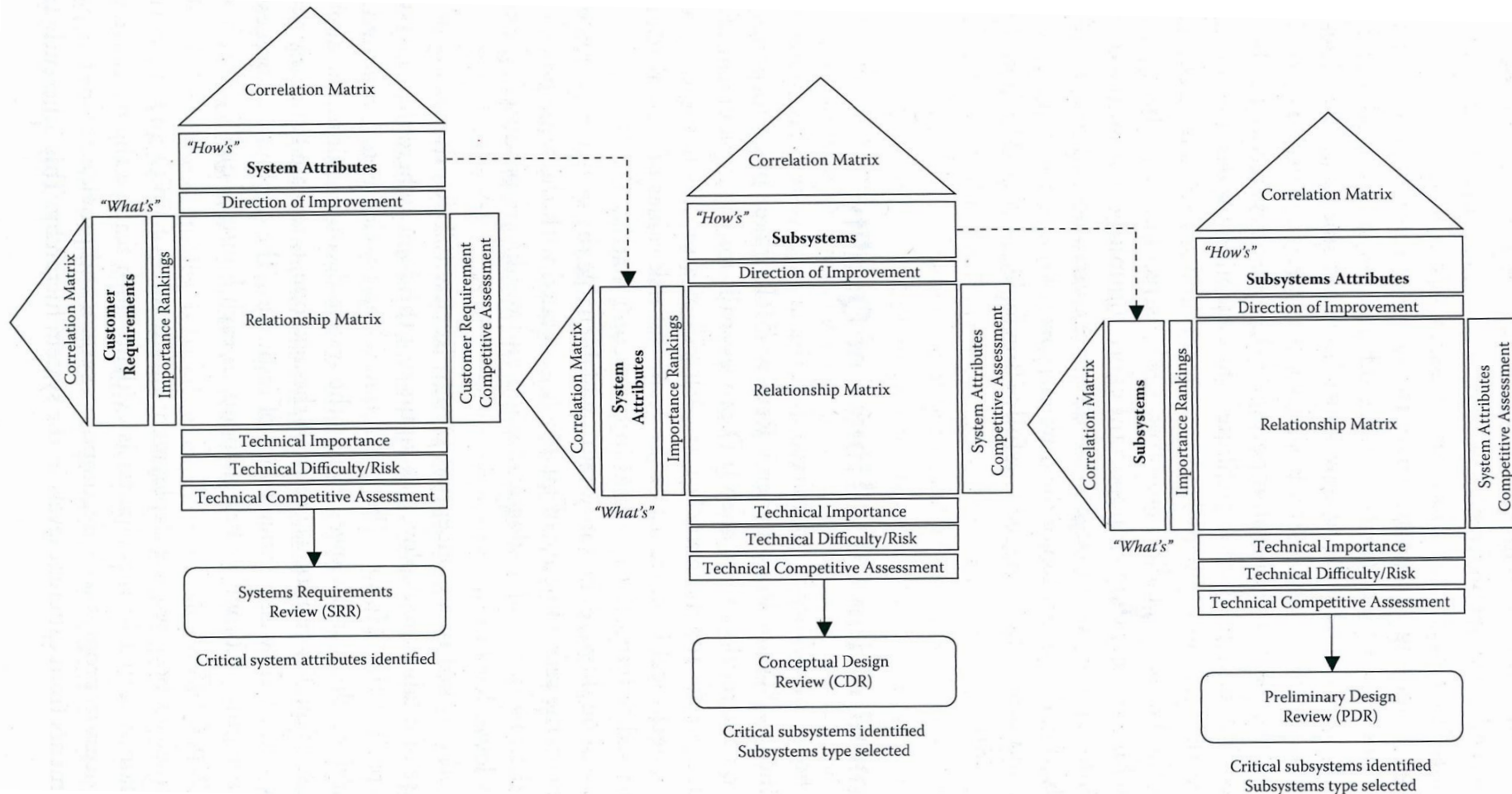
- Set of existing system is identified as competitors.
- Based on experts assessment.

❑ Technical Competitive Assessment.

- Assessment done for competitive systems relative to system attributes.

House of Quality

▣ Levels of decomposition.



Quality Function Deployment

□ Advantages

- Improves the involvement of users and managers.
- Shortens the development lifecycle and improves overall project development.
- Supports team involvement by structuring communication processes.
- Provides a preventive tool that avoids the loss of information.

Quality Function Deployment

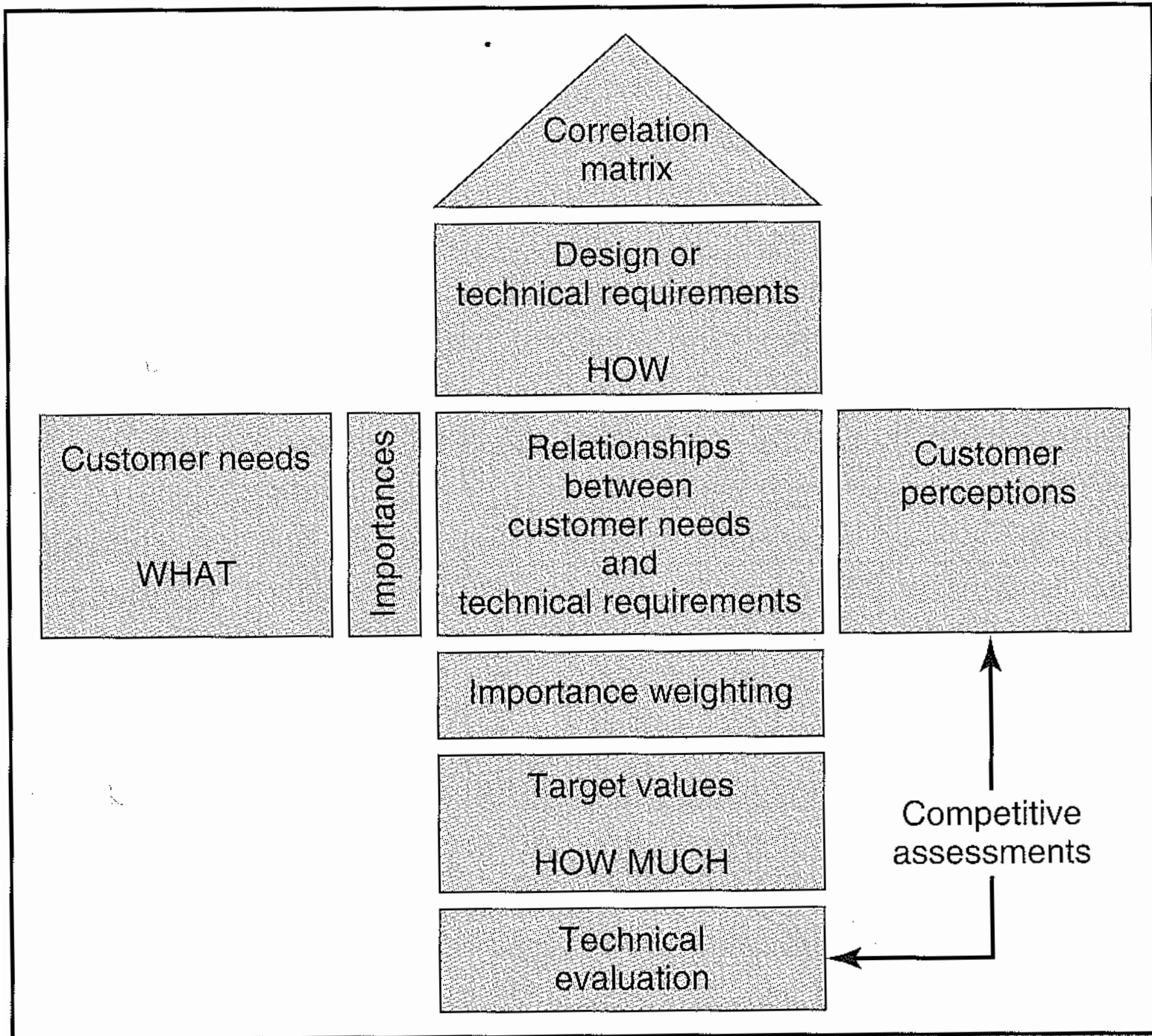
❑ Disadvantages

- Difficulty in expressing temporal requirements.
- Difficulty in use with an entirely new project type.
 - ❑ How do you discover customer requirements for something that does not exist?
 - ❑ How do you build and analyze the competitive products?
- Sometimes it is hard to find measurements for certain functions and to keep the level of abstraction uniform.
- As the feature list grows uncontrollably, the house of quality can become a “mansion”

Quality Function Deployment

- Example:

- House of Quality for a Television Remote Control Switch.



Television Remote Control Switch

□ Rows: Customer Requirements (Needs).

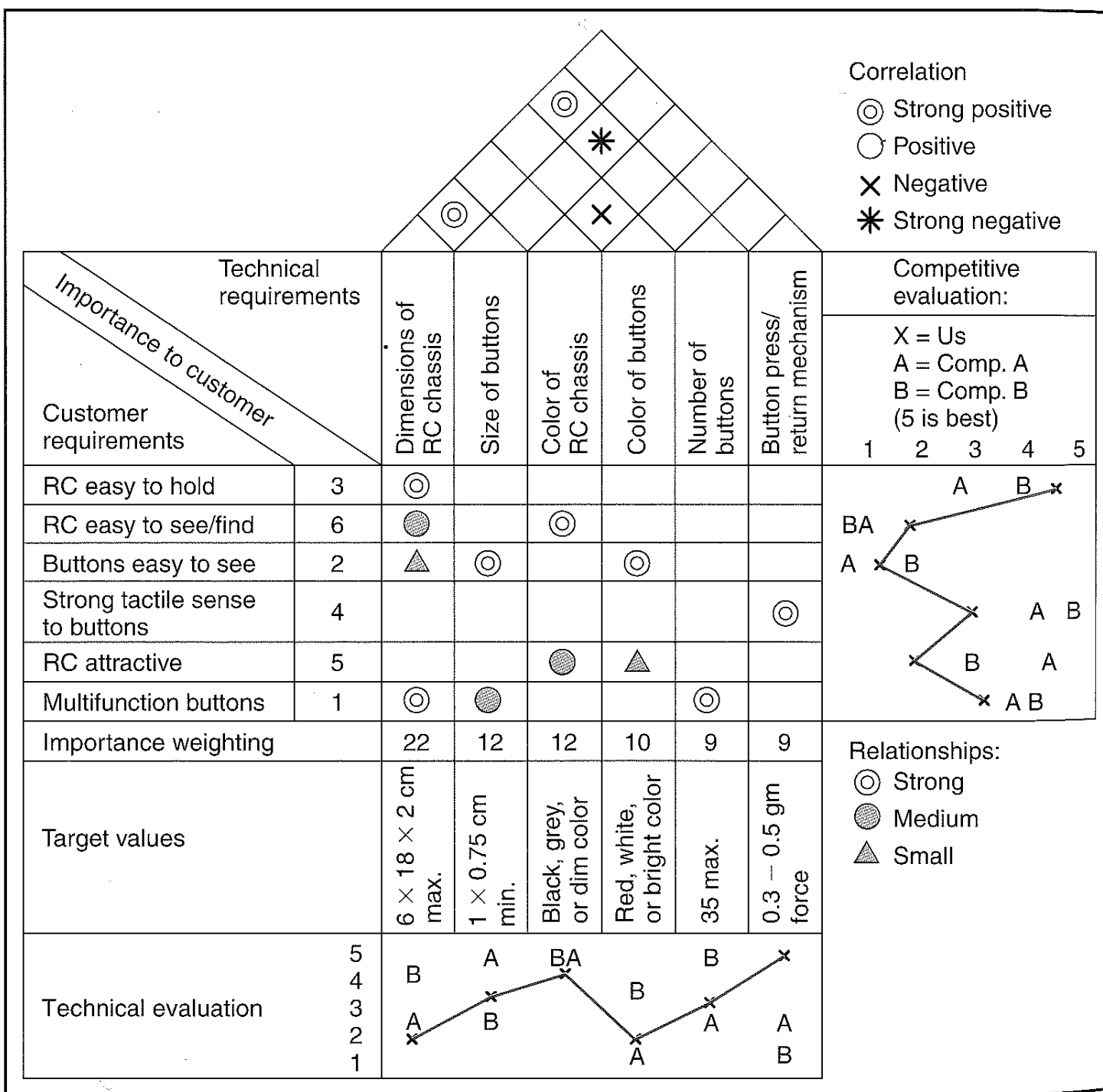
- What the customers think is important for the product.
- Product “what’s”
- Importance to customer: requirements have been rank ordered 1-6 by customer preference.
 - “Multifunction buttons” is rated the highest.
 - “RC easy to see/find” is rated the lowest.

□ Columns: Technical Requirements.

- Requirements or attributes of the product, the ways that the product meets customer requirements.
- Product “how’s”

Television Remote Control Switch

- Central matrix: Relationships between the “what’s” and “how’s”
 - Show the strength of the relationships: strong positive, positive, negative, strong negative.
 - “Buttons easy to use” has a strong positive relationship to the “Size of buttons” and “Color of buttons”
 - “Buttons easy to use” has a positive relationship to the “Dimensions of the RC chassis”
 - Each relationship has a numerical weighting of 1, 3, or 9.



Television Remote Control Switch

□ Importance weighting.

- Weights of the symbols in each column are summed to determine the relative importance of the technical attributes.
- Most important technical attribute is “Dimensions of RC chassis”
 - $\text{Weight} = 9 + 3 + 1 + 9 = 22$
- Next most important attributes is “Size of buttons” and “Color of RC chassis”
 - $\text{Weight} = 9 + 3 = 12$

Television Remote Control Switch

- ❑ **Roof: Correlation matrix among the technical attributes.**
 - “Dimensions of RC chassis” has a strong positive correlation with “Size of buttons” and “Number of buttons”
 - “Size of buttons” has a strong negative correlation with “Number of buttons”
 - ❑ Smaller buttons allow space for more buttons; large buttons, space for fewer.
- ❑ **Target values.**
 - Numerical or qualitative descriptions are design target set for the technical attributes.
 - One target of the design is to keep the “Dimensions of RC chassis” within “6 x 18 x 2”

Television Remote Control Switch

□ Technical evaluation.

- Graph compares the company on the technical attributes.
- X = "us" against two of its competitors, A and B.
- The company's current product does relatively poorly on the attributes of "Dimensions of RC chassis" and "Color of buttons"
- The company's current product fares well on "Color of RC chassis" and "Button press/return mechanism"
- These evaluations are based on test results and opinions of engineers.

Television Remote Control Switch

□ Competitive evaluation.

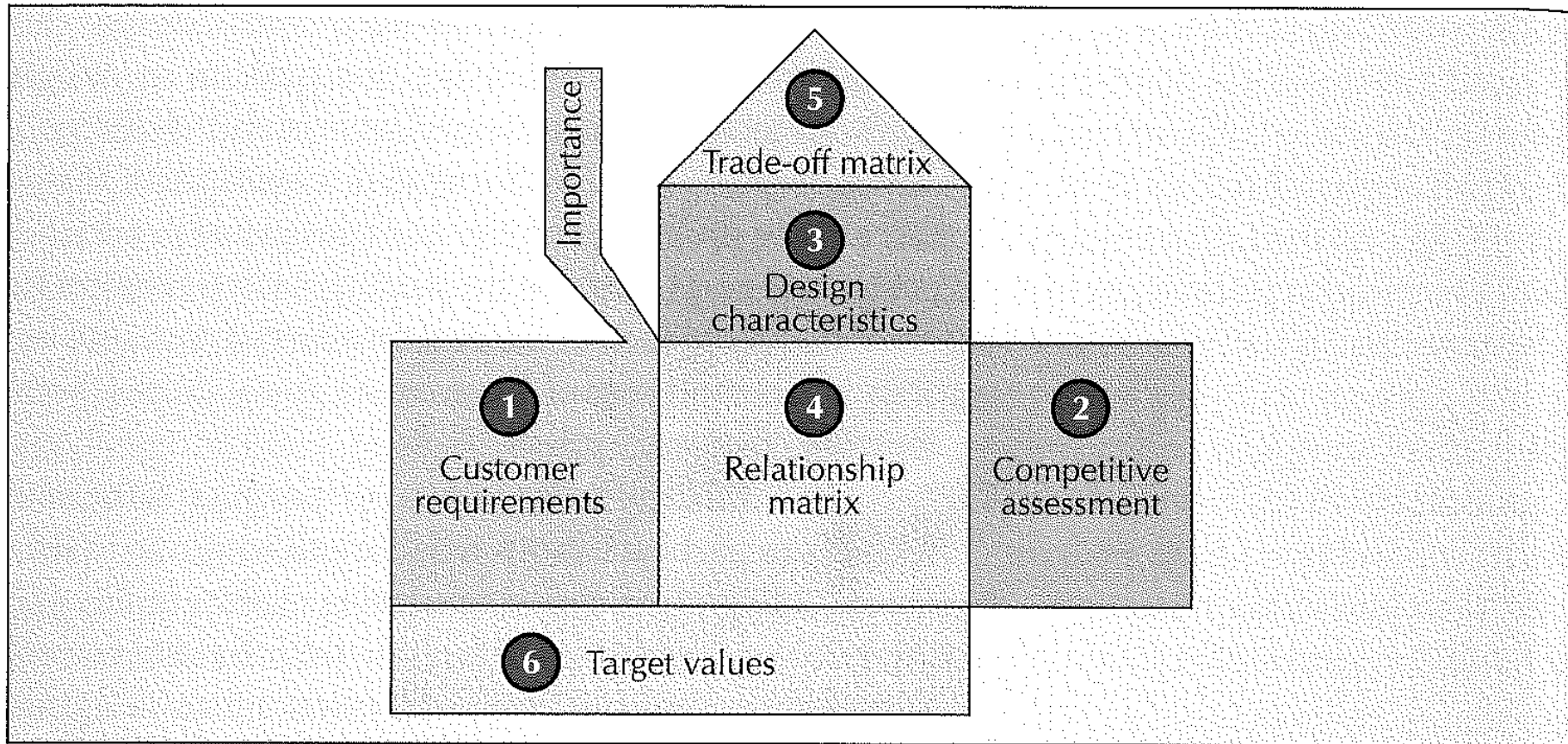
- Graph rates the company and its competitors in terms of customer requirements based on customer surveys.
- Customers think the company does best in “RC easy to hold” but worst in terms of “Buttons easy to see”
- Suggests areas in which designers might focus the effort to gain a market niche.
- Rating indicates that no company does well on “Buttons easy to see” despite the fact that customers rank that requirement second in importance.
- A requirement that customers rank high, yet on which they rank all companies low, suggests a design feature that could be exploited to improve a company’s competitive standing.

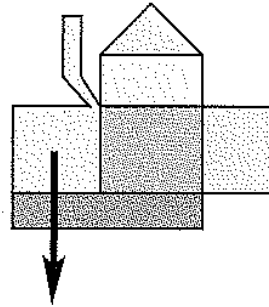
Quality Function Deployment

- Example:

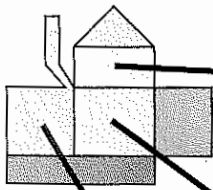
- House of Quality for a Steam Iron.

Steam Iron Example





Customer Requirements			Competitive Assessment				
			1	2	3	4	5
Irons well	Presses quickly	9	B	A		X	
	Removes wrinkles	8		AB			X
	Doesn't stick to fabric	6	X			BA	
	Provides enough steam	8			AB		X
	Doesn't spot fabric	6		X		AB	
	Doesn't scorch fabric	9			A	X	B
Easy and safe to use	Heats quickly	6		X		B	A
	Automatic shut-off	3					ABX
	Quick cool-down	3		X		A	B
	Doesn't break when dropped	5			AB		X
	Doesn't burn when touched	5		AB	X		
	Not too heavy	8		X			A B



Customer Requirements

Design characteristics

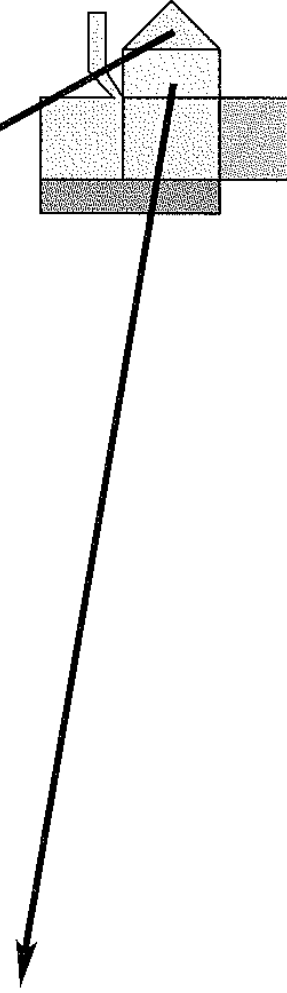
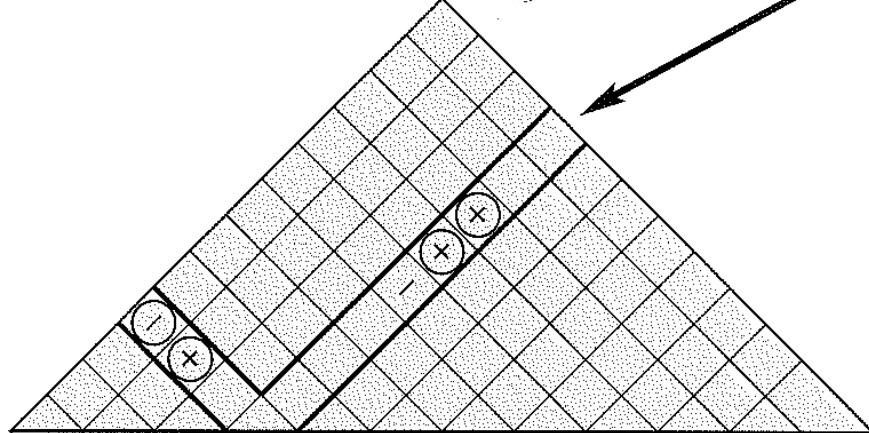
Irons well	Presses quickly	-	(-)	+	+	+				-			
	Removes wrinkles		(+)		+		+	+	+				
	Doesn't stick to fabric		-			(+)			+		(+)	+	
	Provides enough steam			+			+	+	+				
	Doesn't spot fabric					+	-	-	(-)				
	Doesn't scorch fabric				+	(+)			+	-	(+)		
	Easy and safe to use	Heats quickly			-	-					(+)		-
Automatic shut-off											+		(+)
Quick cool-down				-	(-)	+					(+)		
Doesn't break when dropped			+	+	(+)							+	
Doesn't burn when touched					+						+	(+)	+
Not too heavy		+	(-)	-	-	(+)						-	

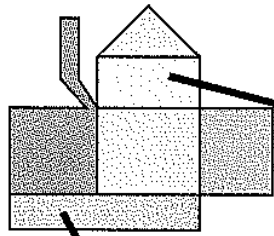
Relationship matrix

Relationships

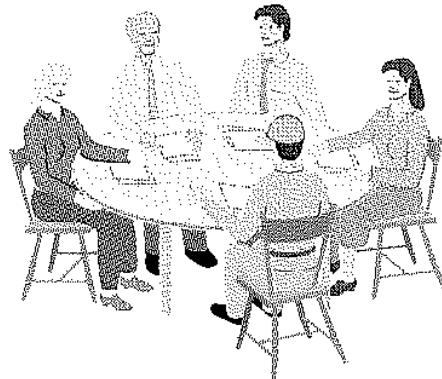
- (+) Strong positive
- + Medium positive
- Medium negative
- (-) Strong negative

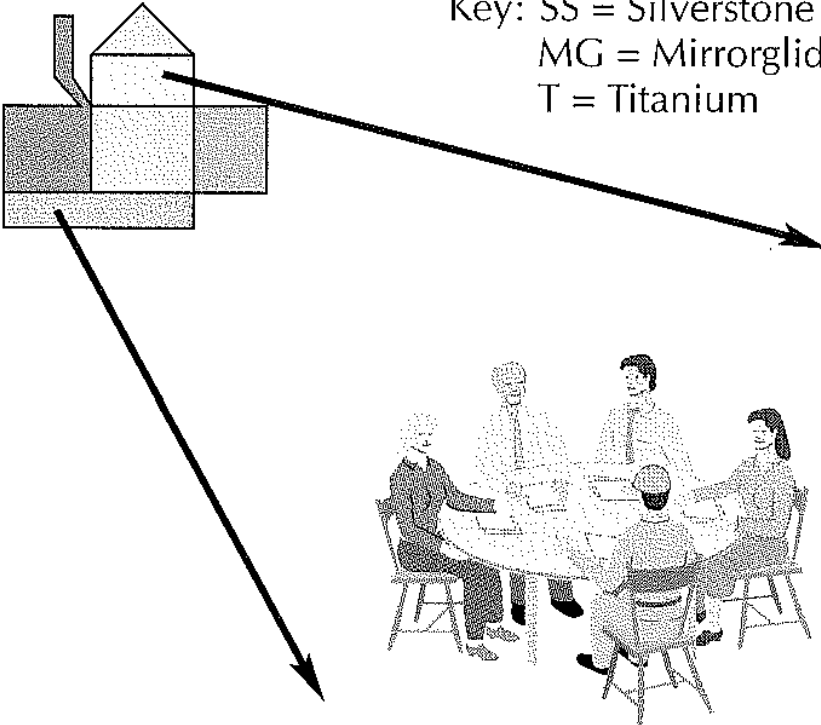
Energy needed to press
Weight of iron
Size of soleplate
Thickness of soleplate
Material used for soleplate
Number of holes
Size of holes
Flow of water from holes
Time required to reach 450° F
Time required to go from 450° to 100° F
Protective cover for soleplate
Automatic shutoff





Key: SS = Silverstone
MG = Mirrorglide
T = Titanium



		Key: SS = Silverstone MG = Mirroglide T = Titanium													
Objective measures	Units of measure	ft-lb	lbs	in.	cm	ty	ea	mm	oz/s	sec	sec	Y/N	Y/N		
	Iron A	3	1.4	8×4	2	SS	27	15	0.5	45	500	N	Y		
	Iron B	4	1.2	8×4	1	MG	27	15	0.3	35	350	N	Y		
	Our Iron (X)	2	1.7	9×5	4	T	35	15	0.7	50	600	N	Y		
Estimated impact		3	4	4	4	5	4	3	2	5	5	3	0		
Estimated cost		3	3	3	3	4	3	3	3	4	4	5	2		
Target values			1.2	8×5	3	SS	30			30	500				
Design changes			*	*	*	*	*			*	*				

		Competitive Assessment												
		1	2	3	4	5								
Customer Requirements	Importance													
		Force	Dimensions & Material		Steam		Time		Safety					
		Energy needed to press	Weight of iron	Size of soleplate	Thickness of soleplate	Material used for soleplate	Number of holes	Size of holes	Flow of water from holes	Time required to reach 450° F	Time required to go from 450° to 100° F	Protective cover for soleplate	Automatic shutoff	
Irons well	Presses quickly	9	-	(-)	+	+	+			-				B A X
	Removes wrinkles	8	(+)		+		+	+	+					AB X
	Doesn't stick to fabric	6	-			(+)			+	(+)	+			X BA
	Provides enough steam	8		+			+	+	+					AB X
	Doesn't spot fabric	6				+	-	-	(-)					X AB
	Doesn't scorch fabric	9			+	(+)		+	-	(+)				A XB
Easy and safe to use	Heats quickly	6		-	-					(+)		-		X B A
	Automatic shut-off	3								(+)		(+)		ABX
	Quick cool-down	3		-	(-)	+				+				X A B
	Doesn't break when dropped	5		+	+	(+)					+			AB X
	Doesn't burn when touched	5				+				+	(+)	+		ABX
	Not too heavy	8	+	(-)	-	(+)						-		X A B
Objective measures	Units of measure	ft-lb	lb	in	cm	ty	ea	mm	oz/8	sec.	sec.	Y/N	Y/N	
	Iron A	3	1.4	8x4	2	SS	27	15	0.5	45	500	N	Y	
	Iron B	4	1.2	8x4	1	MG	27	15	0.3	35	350	N	Y	
	Our Iron (X)	2	1.7	9x5	4	T	35	15	0.7	50	600	N	Y	
Estimated impact		3	4	4	4	5	4	3	2	5	5	3	0	
Estimated cost		3	3	3	3	4	3	3	3	4	4	5	2	
Target values		2	1.2	8x5	3	SS	30	15	0.7	30	500	N	Y	
Design changes			*	*	*	*	*			*	*			

Key:
 SS = Silverstone
 MG = Mirroglide
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